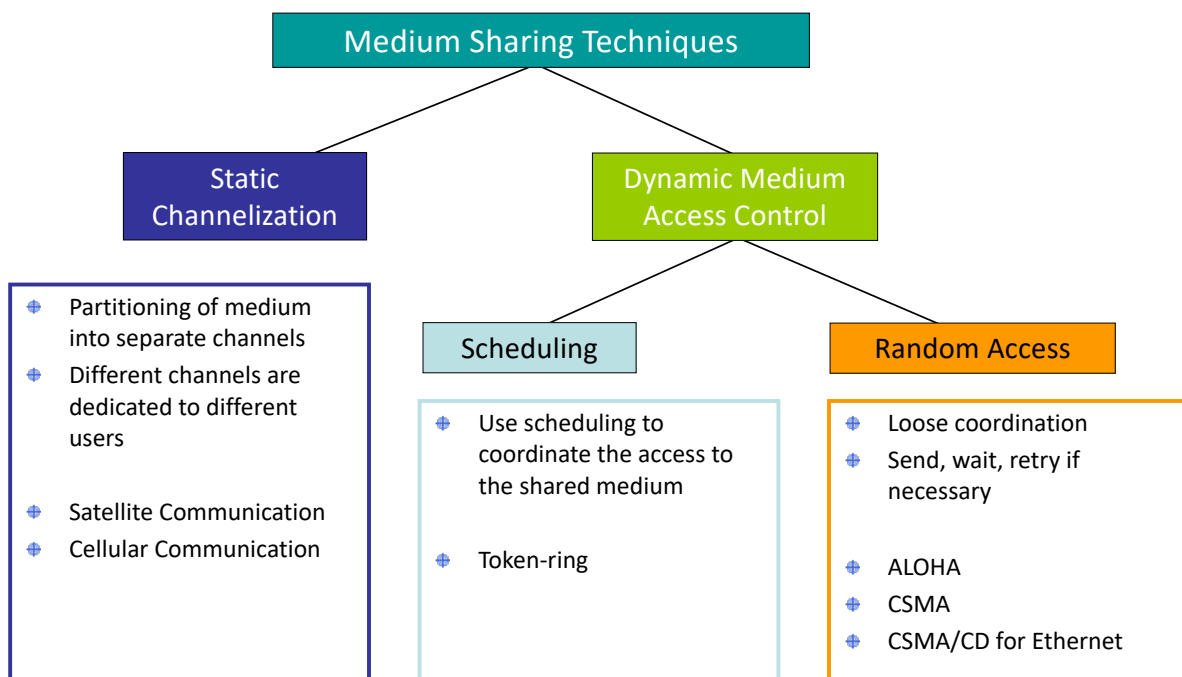


Topic 4: Data Link Layer

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Schemes for Medium Sharing



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CSMA (Carrier Sensing Multiple Access)

- ⊕ A station **senses** the channel before it starts transmission
 - ➡ If idle, start transmission
 - ➡ If busy, either wait or schedule backoff (different options)
 - 1-Persistent CSMA
 - Non-Persistent CSMA

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CSMA/CD (CSMA with Collision Detection)

- ⊕ In CSMA schemes, collisions involve entire frame transmissions
- ⊕ The amount of the wasted bandwidth can be reduced by **aborting the transmission as soon as a collision is detected**
 - ➡ If a collision is detected during the transmission, the station
 - aborts the transmission
 - sends a short jamming signal to ensure that other stations know that a collision has occurred, and
 - use a backoff algorithm to schedule a future re-sensing time

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